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Inventor(s)	Louie; Michael Kenneth-Que

Reusable composite curing system

Abstract

A reusable composite curing system comprising a reusable bladder; an end fitting sealed to the reusable bladder, the end fitting containing a passage in communication with an interior of the reusable bladder and a first half of the pneumatic connector secured within the end fitting and in communication with the passage; and a cure tool having a venting channel extending from a supporting surface of the cure tool and exiting through one of an underside or a side of the cure tool, and a second half of a pneumatic connector secured within the venting channel; the first half of the pneumatic connector releasable from the second half of the pneumatic connector while maintaining the second half of the pneumatic connector in the cure tool and while maintaining the first half of the pneumatic connector in the end fitting.

Inventors:	Louie; Michael Kenneth-Que (Renton, WA)
Applicant:	The Boeing Company (Arlington, VA)
Family ID:	91580942
Assignee:	The Boeing Company (Arlington, VA)
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Primary Examiner: Kim; Yunju

Attorney, Agent or Firm: Yee & Associates, P.C.

Background/Summary

BACKGROUND INFORMATION

1. Field

(1) The present disclosure generally relates to methods and equipment for fabricating composite resin parts, and deals more particularly with a reusable composite curing system used in curing composite parts.

2. Background

(2) Composite resin parts may be cured within an autoclave that applies heat and pressure to the part during a cure cycle. Some part geometries include internal cavities that may cause the part to collapse under autoclave pressure unless a tool such as an inflatable bladder is placed in the cavity. Such an inflatable bladder may be inflated during a cure process so as to react the autoclave pressure force applied to the part. Typically, these inflatable bladders are pressurized by venting them to the autoclave pressure through a vacuum bag.

(3) Venting through the vacuum bag requires additional sealing which can take an undesirable amount of time and generate waste from seals and disposable vacuum bags. Additionally, failure to

properly vent through the vacuum bag can result in the reusable bladder failing to pressurize and undesirably affecting the resulting composite part quality.

(4) Therefore, it would be desirable to have a method and apparatus that takes into account at least some of the issues discussed above, as well as other possible issues.

SUMMARY

(5) An embodiment of the present disclosure provides a reusable composite curing system. The reusable composite curing system comprises a reusable bladder; an end fitting sealed to the reusable bladder; and a cure tool having a venting channel extending from a supporting surface of the cure tool and exiting through one of an underside or a side of the cure tool. The end fitting contains a passage in communication with an interior of the reusable bladder and a first half of the pneumatic connector secured within the end fitting and in communication with the passage. The cure tool has a second half of a pneumatic connector secured within the venting channel. The first half of the pneumatic connector is releasable from the second half of the pneumatic connector while maintaining the second half of the pneumatic connector in the cure tool and while maintaining the first half of the pneumatic connector in the end fitting.

(6) Another embodiment of the present disclosure provides a reusable composite curing system. The reusable composite curing system comprises a reusable bladder configured to apply pressure to an internal cavity of a composite part during autoclave curing, an end fitting sealed to the reusable bladder, the end fitting containing a passage in communication with an interior of the reusable bladder and a first half of a pneumatic connector in communication with the passage and secured within the end fitting, and a reusable vacuum bag sealed over the reusable bladder and the end fitting. The first half of the pneumatic connector is releasable from a second half of the pneumatic connector while maintaining the half of the pneumatic connector in the end fitting.

(7) Another embodiment of the present disclosure provides a method of curing a composite material. A reusable bladder is lowered towards a cure tool such that a first half of a pneumatic connector secured in an end fitting sealed to the reusable bladder connects to a second half of a pneumatic connector secured in the cure tool, the pneumatic connector forming a pressure passageway to provide an external pressure to an interior of the reusable bladder. Heat and pressure are applied to the composite material supported on the cure tool and the reusable bladder to cure the composite material to form a composite part having an internal cavity.

(8) The features and functions can be achieved independently in various embodiments of the present disclosure or may be combined in yet other embodiments in which further details can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The novel features believed characteristic of the illustrative embodiments are set forth in the appended claims. The illustrative embodiments, however, as well as a preferred mode of use, further objectives and features thereof, will best be understood by reference to the following detailed description of an illustrative embodiment of the present disclosure when read in conjunction with the accompanying drawings, wherein:

(2) FIG. 1 is an illustration of an aircraft in accordance with an illustrative embodiment;

(3) FIG. 2 is an illustration of a block diagram of a manufacturing environment in accordance with an illustrative embodiment;

(4) FIG. 3 is an isometric view of the reusable composite curing system with some portions in phantom in accordance with an illustrative embodiment;

(5) FIG. 4 is an exploded cross-sectional view of the reusable composite curing system with some portions in phantom in accordance with an illustrative embodiment;

- (6) FIG. 5 is an isometric view of the reusable composite curing system supporting composite material for curing in accordance with an illustrative embodiment;
- (7) FIG. 6 is a cross-sectional view of the reusable composite curing system during curing of composite material in accordance with an illustrative embodiment;
- (8) FIG. 7 is a flowchart of a method of curing a composite material in accordance with an illustrative embodiment;
- (9) FIG. 8 is an illustration of an aircraft manufacturing and service method in a form of a block diagram in accordance with an illustrative embodiment; and
- (10) FIG. 9 is an illustration of an aircraft in a form of a block diagram in which an illustrative embodiment may be implemented.

DETAILED DESCRIPTION

- (11) Turning now to FIG. 1, an illustration of an aircraft is depicted in accordance with an illustrative embodiment. Aircraft **100** has wing **102** and wing **104** attached to body **106**. Aircraft **100** includes engine **108** attached to wing **102** and engine **110** attached to wing **104**.
- (12) Body **106** has tail section **112**. Horizontal stabilizer **114**, horizontal stabilizer **116**, and vertical stabilizer **118** are attached to tail section **112** of body **106**.
- (13) Aircraft **100** is an example of an aircraft that can have composite parts formed using the reusable composite curing system of the illustrative examples. For example, at least one of wing **102**, wing **104**, or body **106** can have a skin and stiffeners cured using the reusable composite curing system of the illustrative examples.
- (14) Turning now to FIG. 2, an illustration of a block diagram of a manufacturing environment is depicted in accordance with an illustrative embodiment. Reusable composite curing system **202** can be used to cure components of aircraft **100** of FIG. 1. Reusable composite curing system **202** is used within manufacturing environment **200** to form composite part **204** having internal cavity **206**.
- (15) Reusable composite curing system **202** comprises reusable bladder **208**, end fitting **210** sealed to reusable bladder **208**, and cure tool **211** having venting channel **212** extending from supporting surface **214** of cure tool **211** and exiting through one of underside **216** or a side of cure tool **211**, such as side **218** or side **220**. End fitting **210** contains passage **222** in communication with interior **224** of reusable bladder **208** and first half **226** of pneumatic connector **228** secured within end fitting **210** and in communication with passage **222**. Second half **230** of pneumatic connector **228** is secured within venting channel **212**. First half **226** of pneumatic connector **228** is releasable from second half **230** of pneumatic connector **228** while maintaining second half **230** of pneumatic connector **228** in cure tool **211** and while maintaining first half **226** of pneumatic connector **228** in end fitting **210**.
- (16) First half **226** takes the form of any desirable type of pneumatic connector half. In some illustrative examples, pneumatic connector **228** is a quick connector. In these illustrative examples, first half **226** can be one of a male quick connector or a female quick connector.
- (17) Second half **230** takes the form of any desirable type of pneumatic connector half. In some illustrative examples, pneumatic connector **228** is a quick connector. In these illustrative examples, second half **230** can be one of a male quick connector or a female quick connector. In some illustrative examples, one of first half **226** and second half **230** of pneumatic connector **228** is a female quick connector, and the other of first half **226** and second half **230** is a male quick connector.
- (18) As depicted, wherein end fitting **210** comprises threaded bore **232** extending through underside **234** of end fitting **210**. Threaded bore **232** does not extend through the thickness of end fitting **210**. Threaded bore **232** terminates into passage **222**. First half **226** of pneumatic connector **228** is secured within threaded bore **232**. In some illustrative examples, first half **226** remains secured within end fitting **210** during and between curing operations.
- (19) Passage **222** extends into end fitting **210** in a manner to pneumatically connect to first half **226** of pneumatic connector **228**. In some illustrative examples, passage **222** runs perpendicular to first

half **226** of pneumatic connector **228**. In some illustrative examples, passage **222** runs parallel to a longitudinal axis of end fitting **210**. In some illustrative examples, passage **222** runs perpendicular to threaded bore **232**.

(20) Pressure passageway **236** provides access to interior **224** of reusable bladder **208** for pressurization. In some illustrative examples, passage **222**, pneumatic connector **228**, and venting channel **212** define pressure passageway **236** that allows an interior of the reusable bladder to be subjected to external pressure **238**. End fitting **210** is formed of material **239**.

(21) Material **239** can be any desirable material configured to withstand the heat and pressure of autoclave **240**. End fitting **210** is formed of any desirable material configured to maintain its shape when subjected to increased temperature and increased pressure during composite curing. In some illustrative examples, end fitting **210** is formed of metallic material **242**.

(22) Material **239** is selected that has material characteristics **244** to withstand curing within autoclave **240**. Material **239** is selected that has material characteristics **244** to remain pneumatically sealed to reduce or prevent leaks between end fitting **210** and pneumatic connector **228**. Material **239** is selected that has material characteristics **244** that can be sealed to reusable bladder **208**.

(23) To secure end fitting **210** to reusable bladder **208**, sealant **246** is applied between reusable bladder **208** and end fitting **210**. Material **239** can be selected to seal to reusable bladder **208** using sealant **246**.

(24) In some illustrative examples, material **239** is machined to form end fitting **210**. For example, metallic materials can be machined to a desirable shape for end fitting **210**. In some illustrative examples, material **239** for end fitting **210** can be printed or molded.

(25) Reusable composite curing system **202** comprises reusable vacuum bag **248** sealed to cure tool **211** and completely covering reusable bladder **208** and end fitting **210**. Reusable vacuum bag **248** is sealed to cure tool **211** using seal **250**. Seal **250** surrounds a perimeter of reusable vacuum bag **248**. Seal **250** forms a sealed volume between reusable vacuum bag **248** and cure tool **211**. The sealed volume between reusable vacuum bag **248** and cure tool **211** can be referred to as vacuum chamber **256**. Seal **250** can take any desirable disposable or reusable form.

(26) After sealing reusable vacuum bag **248** to cure tool **211**, composite material **252** and composite material **254** can be cured to form composite part **204** having internal cavity **206**. Composite material **252** is applied to cure tool **211** prior to connecting end fitting **210** to cure tool **211** using pneumatic connector **228**. Although supporting surface **214** appears planar, supporting surface **214** can have any desirable geometry. In some illustrative examples, when supporting surface **214** is substantially planar, composite material **252** can be cured into a composite skin. In some illustrative examples, when supporting surface **214** has a trough or other geometric feature, composite material **252** can be cured to form a stiffener with internal cavity **206**.

(27) Composite material **254** can be cured to form either a composite skin or a composite stiffener, depending upon the design of cure tool **211**. Reusable bladder **208** is configured to apply pressure to internal cavity **206** of composite part **204** during autoclave curing. The pressure applied by reusable bladder **208** is greater than a pressure under the reusable vacuum bag **248**. The geometry of cure tool **211** and reusable bladder **208** can be modified to form composite part **204** having any desirable design with internal cavity **206**.

(28) Reusable composite curing system **202** and methods of use enable use of reusable vacuum bag **248** for composite curing. The illustrative examples enable high rate manufacturing. Using reusable vacuum bag **248** reduces time sealing the vacuum bag. Utilizing pressure passageway **236** to pressurize interior **224** of reusable bladder **208** enables use of reusable vacuum bag **248**. The illustrative examples increase sustainability of composite curing. The illustrative examples provide pneumatic connector **228** in end fitting **210** and cure tool **211**. When applying reusable bladder **208** to cure tool **211**, pneumatic connector **228** indexes and quickly connects. Use of pneumatic connector **228** to provide pressurization to reusable bladder **208** can provide at least one of reduced

preparation time for sealing reusable vacuum bag **248** to cure tool **211**, reduced cost, or increased sustainability by using fewer disposable resources.

(29) Pneumatic connector **228** provides connection between reusable bladder **208** and cure tool **211** quickly and without any additional tooling. Pneumatic connector **228** can also provide indexing of reusable bladder **208** relative to cure tool **211**.

(30) The illustration of manufacturing environment **200** in FIG. 2 is not meant to imply physical or architectural limitations to the manner in which an illustrative embodiment may be implemented. Other components in addition to or in place of the ones illustrated may be used. Some components may be unnecessary. Also, the blocks are presented to illustrate some functional components. One or more of these blocks may be combined, divided, or combined and divided into different blocks when implemented in an illustrative embodiment.

(31) For example, in some illustrative examples, multiple seals can be used to seal reusable vacuum bag **248** to cure tool **211** instead of a single seal, seal **250**. In some illustrative examples, seal **250** is a first part of a interlocking seal and a second part (not depicted) of the interlocking seal is connected to the reusable vacuum bag **248**.

(32) Turning now to FIG. 3, an isometric view of the reusable composite curing system with some portions in phantom is depicted in accordance with an illustrative embodiment. Reusable composite curing system **300** is a physical implementation of reusable composite curing system **202** of FIG. 2.

(33) Reusable composite curing system **300** comprises reusable bladder **302**, end fitting **304** sealed to reusable bladder **302**, and cure tool **316** having venting channel **314** extending from supporting surface **318** of cure tool **316** and exiting through underside **320** of cure tool **316**. End fitting **304** comprises passage **306** in communication with interior **303** of reusable bladder **302**. End fitting **304** further comprises first half **310** of pneumatic connector **311** secured within end fitting **304** and in communication with passage **306**. In this illustrative example, first half **310** of pneumatic connector **311** is secured within threaded bore **308**. Threaded bore **308** is in communication with passage **306**.

(34) Second half **312** of pneumatic connector **311** is secured within venting channel **314**. First half **310** of pneumatic connector **311** is releasable from second half **312** of pneumatic connector **311** while maintaining second half **312** of pneumatic connector **311** in cure tool **316** and first half **310** of pneumatic connector **311** in end fitting **304**.

(35) In this illustrative example, first half **310** of pneumatic connector **311** is a female quick connector. First half **310** of pneumatic connector **311** can take any desirable form. In some other illustrative examples, first half **310** of pneumatic connector **311** is a male connector. In some other illustrative examples, first half **310** of pneumatic connector **311** can have a different female connector design than that depicted.

(36) In this illustrative example, second half **312** of pneumatic connector **311** is a male quick connector. Second half **312** of pneumatic connector **311** can take any desirable form. In some other illustrative examples, second half **312** of pneumatic connector **311** is a female connector. In some other illustrative examples, second half **312** of pneumatic connector **311** can have a different male connector design than that depicted.

(37) To apply external pressure to interior **303** of reusable bladder **302** during curing of a composite material, first half **310** of pneumatic connector **311** is connected to second half **312** of pneumatic connector **311**. By connecting first half **310** to second half **312**, a pressure passageway is formed. The pressure passageway is formed by passage **306**, threaded bore **308**, first half **310** of pneumatic connector **311**, second half **312** of pneumatic connector **311**, and venting channel **314**.

(38) The illustration of reusable composite curing system **300** in FIG. 3 is not meant to imply physical or architectural limitations to the manner in which an illustrative embodiment may be implemented. Although reusable bladder **302** and end fitting **304** have a trapezoidal cross-section, reusable bladder **302** and end fitting **304** can have any desirable geometry.

(39) Additionally, although cure tool **316** is depicted as having a planar forming surface **318**, in other illustrative examples, a trough can be present in cure tool **316**. Although reusable bladder **302**

is depicted as resting atop cure tool **316**, in other illustrative examples, reusable bladder **302** can be placed over composite material within a trough of cure tool **316**.

(40) Turning now to FIG. **4**, an exploded cross-sectional view of the reusable composite curing system with some portions in phantom is depicted in accordance with an illustrative embodiment. View **400** is an exploded cross-sectional view of reusable composite curing system **300** of FIG. **3**. In view **400**, first half **310** of pneumatic connector **311** is exploded out of threaded bore **308** for a better view of threaded bore **308**. During positioning and operation, first half **310** of pneumatic connector **311** remains secured within threaded bore **308**.

(41) In view **400**, second half **312** of pneumatic connector **311** is exploded out of venting channel **314** for a better view of venting channel **314**. During positioning and operation, second half **312** of pneumatic connector **311** remains secured within venting channel **314**.

(42) As depicted, reusable bladder **302** is sealed to exterior surface **402** of end fitting **304**. As depicted, by reusable bladder **302** being sealed to exterior surface **402** of end fitting **304**, a portion of end fitting **304** extends into interior **303** of reusable bladder **302**.

(43) Turning now to FIG. **5**, an isometric view of the reusable composite curing system supporting composite material for curing is depicted in accordance with an illustrative embodiment. View **500** is an isometric view of reusable composite curing system **501** supporting composite material for curing. Reusable composite curing system **501** comprises cure tool **502** and reusable bladder **506** connected to end fitting **504**. In this illustrative example, composite material **508** is supported by cure tool **502**. Composite material **508** takes the form of a composite skin. Composite material **510** is supported by reusable bladder **506**. Composite material **510** forms a stiffener with an internal cavity.

(44) Turning now to FIG. **6**, a cross-sectional view of the reusable composite curing system during curing of composite material is depicted in accordance with an illustrative embodiment. View **600** is a cross-sectional view of reusable composite curing system **601** during curing of a composite material. In view **600**, autoclave **602** is used to cure composite material **618** and composite material **620**. Reusable composite curing system **601** comprises reusable bladder **604** sealed to end fitting **606** and cure tool **608**. Reusable bladder **604** supports composite material **620**. Cure tool **608** supports composite material **618**.

(45) End fitting **606** interfaces with cure tool **608** to allow pressurization of the interior of reusable bladder **604**. Reusable vacuum bag **610** is sealed to cure tool **608** by seal **614** and seal **616**. Reusable bladder **604** is pressurized by pressure **612** inside autoclave **602** without venting through reusable vacuum bag **610**. Reusable bladder **604** is pressured by pressure **612** inside autoclave **602** through pressure passageway **621**. Pressure passageway **621** allows the interior of reusable bladder **604** to be subjected to pressure **612**.

(46) Pressure passageway **621** is defined by passage **622**, pneumatic connector **627**, and venting channel **624**. Venting channel **624** extends through cure tool **608**. Second half **626** of pneumatic connector **627** is secured within venting channel **624**. First half **628** of pneumatic connector **627** is secured within threaded bore **630**. Passage **622** runs parallel to reusable bladder **604** into end fitting **606**. Passage **622** connects to threaded bore **630** in end fitting **606**. Passage **622** is perpendicular to first half **628** of pneumatic connector **627**. As depicted, first half **628** of pneumatic connector **627** is a female connector while second half **626** of pneumatic connector **627** is a male connector.

(47) Although venting channel **624** is depicted as extending through supporting surface **632** to underside **634**, in other illustrative examples a venting channel can extend through side **636** or side **638** to allow pressure **612** to access the interior of reusable bladder **604**. In some illustrative examples, pressure **612** is referred to as an external pressure as pressure **612** is outside a vacuum chamber formed between reusable vacuum bag **610** and cure tool **608**.

(48) Turning now to FIG. **7**, a flowchart of a method of curing a composite material is depicted in accordance with an illustrative embodiment. Method **700** can be used to form components of aircraft **100** of FIG. **1**. Method **700** can be performed using reusable composite curing system **202**

of FIG. 2. Method **700** can be performed using reusable composite curing system **300** of FIGS. 3 and 4. Method **700** can be performed using reusable composite curing system **501** of FIG. 5. Method **700** can be performed using reusable composite curing system **601** of FIG. 6.

(49) Method **700** lowers a reusable bladder towards a cure tool such that a first half of a pneumatic connector secured in an end fitting sealed to the reusable bladder connects to a second half of a pneumatic connector secured in the cure tool, the pneumatic connector forming a pressure passageway to provide an external pressure to an interior of the reusable bladder (operation **702**). The first half of the pneumatic connector is secured within the end fitting prior to lowering the reusable bladder. In some illustrative examples, method **700** secures the first half of the pneumatic connector within the end fitting prior to lowering the reusable bladder. The end fitting is sealed to the reusable bladder prior to lowering the reusable bladder. In some illustrative examples, method **700** seals the end fitting to the reusable bladder. As a result, the reusable bladder with the end fitting and first half of pneumatic connector is lowered as a single component towards the cure tool. The second half of the pneumatic connector is secured to the cure tool prior to lowering the reusable bladder towards the cure tool. In some illustrative examples, method **700** secures the second half of the pneumatic connector to the cure tool prior to lowering the reusable bladder. By lowering the reusable bladder towards the cure tool, the first half of the second pneumatic connector connects to the second half of the pneumatic connector. In some illustrative examples, the pneumatic connector is a quick connector.

(50) Method **700** cures the composite material to form a composite part having an internal cavity by applying heat and pressure to the composite material supported on the cure tool and the reusable bladder (operation **704**).

(51) In some illustrative examples, method **700** wraps the composite material around the reusable bladder (operation **706**). In some illustrative examples, the reusable bladder forms an internal cavity for a resulting composite part. In some illustrative examples, lowering the reusable bladder comprises placing the composite material and reusable bladder onto the cure tool (operation **708**).

(52) In some illustrative examples, method **700** places the composite material onto the cure tool (operation **710**). In some illustrative examples, the composite material placed onto the cure tool is a composite skin in a resulting composite part. In some illustrative examples, the composite material placed onto the cure tool is placed into a trough to form a stiffener in a resulting composite part. In some illustrative examples, lowering the reusable bladder towards the cure tool comprises positioning the reusable bladder onto the composite material on the cure tool (operation **712**).

(53) In some illustrative examples, method **700** applies the composite material over the reusable bladder on the cure tool after lowering the reusable bladder towards the cure tool (operation **714**). In some illustrative examples, the composite material applied over the reusable bladder forms a composite skin in a resulting composite part.

(54) In some illustrative examples, method **700** seals a reusable vacuum bag to the cure tool and over the end fitting and reusable bladder (operation **716**). Vents are not placed into reusable vacuum bag for pressurizing the reusable bladder. Sealing the reusable vacuum bag forms a vacuum chamber between the reusable vacuum bag and the cure tool. The composite material to be cured is positioned within the vacuum chamber.

(55) In some illustrative examples, method **700** pressurizes the reusable bladder by subjecting the interior of the bladder to the external pressure by way of the pressure passageway (operation **718**). The external pressure is a pressure outside of the vacuum chamber. When the reusable composite curing system is present in an autoclave, the external pressure is the pressure inside the autoclave. In other illustrative examples, a pressure supply, such as a pump or other pressure source, is connected to the cure tool to apply pressure to the interior of the reusable bladder.

(56) In some illustrative examples, method **700** removes the reusable bladder from the internal cavity of the composite part (operation **720**). In some illustrative examples, method **700** cures a second composite material to form a second composite part using the reusable bladder and the cure

tool (operation **722**). In these illustrative examples, after removal from the internal cavity of the composite part, the reusable bladder and the cure tool are repeatedly used to cure composite parts. (57) As used herein, the phrase “at least one of,” when used with a list of items, means different combinations of one or more of the listed items may be used and only one of each item in the list may be needed. For example, “at least one of item A, item B, or item C” may include, without limitation, item A, item A and item B, or item B. This example also may include item A, item B, and item C or item B and item C. Of course, any combinations of these items may be present. In other examples, “at least one of” may be, for example, without limitation, two of item A; one of item B; and ten of item C; four of item B and seven of item C; or other suitable combinations. The item may be a particular object, thing, or a category. In other words, at least one of means any combination items and number of items may be used from the list but not all of the items in the list are required.

(58) As used herein, “a number of,” when used with reference to items means one or more items.

(59) The flowcharts and block diagrams in the different depicted embodiments illustrate the architecture, functionality, and operation of some possible implementations of apparatuses and methods in an illustrative embodiment. In this regard, each block in the flowcharts or block diagrams may represent at least one of a module, a segment, a function, or a portion of an operation or step.

(60) In some alternative implementations of an illustrative embodiment, the function or functions noted in the blocks may occur out of the order noted in the figures. For example, in some cases, two blocks shown in succession may be executed substantially concurrently, or the blocks may sometimes be performed in the reverse order, depending upon the functionality involved. Also, other blocks may be added in addition to the illustrated blocks in a flowchart or block diagram. Some blocks may be optional. For example, operation **706** through operation **722** may be optional.

(61) Illustrative embodiments of the present disclosure may be described in the context of aircraft manufacturing and service method **800** as shown in FIG. **8** and aircraft **900** as shown in FIG. **9**. Turning first to FIG. **8**, an illustration of an aircraft manufacturing and service method in a form of a block diagram is depicted in accordance with an illustrative embodiment. During pre-production, aircraft manufacturing and service method **800** may include specification and design **802** of aircraft **900** in FIG. **9** and material procurement **804**.

(62) During production, component and subassembly manufacturing **806** and system integration **808** of aircraft **900** takes place. Thereafter, aircraft **900** may go through certification and delivery **810** in order to be placed in service **812**. While in service **812** by a customer, aircraft **900** is scheduled for routine maintenance and service **814**, which may include modification, reconfiguration, refurbishment, or other maintenance and service.

(63) Each of the processes of aircraft manufacturing and service method **800** may be performed or carried out by a system integrator, a third party, and/or an operator. In these examples, the operator may be a customer. For the purposes of this description, a system integrator may include, without limitation, any number of aircraft manufacturers and major-system subcontractors; a third party may include, without limitation, any number of vendors, subcontractors, and suppliers; and an operator may be an airline, a leasing company, a military entity, a service organization, and so on.

(64) With reference now to FIG. **9**, an illustration of an aircraft in a form of a block diagram is depicted in which an illustrative embodiment may be implemented. In this example, aircraft **900** is produced by aircraft manufacturing and service method **800** of FIG. **8** and may include airframe **902** with plurality of systems **904** and interior **906**. Examples of systems **904** include one or more of propulsion system **908**, electrical system **910**, hydraulic system **912**, and environmental system **914**. Any number of other systems may be included.

(65) Apparatuses and methods embodied herein may be employed during at least one of the stages of aircraft manufacturing and service method **800**. One or more illustrative embodiments may be manufactured or used during at least one of component and subassembly manufacturing **806**,

system integration **808**, in service **812**, or maintenance and service **814** of FIG. **8**.

(66) The illustrative examples provide a reusable composite curing system and methods that enable use of reusable vacuum bagging for composite curing. The illustrative examples enable high rate manufacturing of composite curing by reducing time sealing the vacuum bag. The illustrative examples increase sustainability of composite curing. The illustrative examples provide a pneumatic connector in a reusable bladder and a cure tool. When applying the reusable bladder to the cure tool, the pneumatic connector indexes and quickly connects.

(67) The illustrative examples enable the use of reusable vacuum bagging that improves rate and sustainability, saves time, and adds to process robustness. The connection between the pneumatic connector is more robust than tacky tape. The illustrative examples utilize metallic end fittings on reusable bladders. In some illustrative examples, the cure tool comprises a through-hole venting channel. In some illustrative examples, the through-hole venting channel can be created through simple machining.

(68) Quick disconnects are installed into the venting channel of the cure tool and the threaded bore of the end fitting. The threaded bore extends partially into the end fitting. The threaded bore pneumatically connects to a passage extending away from the threaded bore.

(69) The description of the different illustrative embodiments has been presented for purposes of illustration and description, and is not intended to be exhaustive or limited to the embodiments in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different illustrative embodiments may provide different features as compared to other illustrative embodiments. The embodiment or embodiments selected are chosen and described in order to best explain the principles of the embodiments, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. A reusable composite curing system that comprises: a reusable bladder; an end fitting sealed to the reusable bladder, wherein the end fitting comprises: a bore that is threaded, penetrates partially into the end fitting, and connects, at an interior most point of the bore, to a passage that exits the bore perpendicular to an axis of the bore and into communication with an interior of the reusable bladder; and a first half of a pneumatic connector threaded into the bore and in communication with the passage; and a cure tool that comprises a venting channel that extends from a supporting surface of the cure tool and exits through one of an underside or a side of the cure tool; and a second half of the pneumatic connector secured within and extending out from the venting channel, wherein the first half of the pneumatic connector is configured to release from the second half of the pneumatic connector while the second half of the pneumatic connector remains in the cure tool and while the first half of the pneumatic connector remains in the end fitting.
2. The reusable composite curing system of claim 1, wherein the end fitting is formed of a metallic material.
3. The reusable composite curing system of claim 1, wherein one of the first half and the second half of the pneumatic connector is a female quick connector, and wherein the other of the first half and the second half is a male quick connector.
4. The reusable composite curing system of claim 1, wherein the passage runs perpendicular to the first half of the pneumatic connector.
5. The reusable composite curing system of claim 1 further comprising: a reusable vacuum bag sealed to the cure tool, wherein the reusable vacuum bag completely covers the reusable bladder and the end fitting.
6. The reusable composite curing system of claim 1, wherein the passage, pneumatic connector, and venting channel define a pressure passageway that allows the interior of the reusable bladder to be

subjected to an external pressure.

7. The reusable composite curing system of claim 1, wherein threads in the bore in the end fitting stop short of the passage.

8. A reusable composite curing system that comprises: a reusable bladder configured to apply pressure to an internal cavity of a composite part during autoclave curing; an end fitting sealed to the reusable bladder, wherein the end fitting comprises: a bore that is threaded, penetrates partially into the end fitting, and connects, at an interior most point of the bore, to a passage that exits the bore perpendicular to an axis of the bore and into communication with an interior of the reusable bladder; and a first half of a pneumatic connector threaded into the bore and in communication with the passage; a cure tool that comprises a venting channel that extends from a supporting surface of the cure tool and exits through one of an underside or a side of the cure tool; a second half of the pneumatic connector secured within and extending out from the venting channel, wherein the first half of the pneumatic connector is configured to release from the second half of the pneumatic connector while the second half of the pneumatic connector remains in the cure tool and while the first half of the pneumatic connector remains in the end fitting; and a reusable vacuum bag sealed over the reusable bladder and the end fitting, the pressure applied by the reusable bladder greater than a pressure under the reusable vacuum bag.

9. The reusable composite curing system of claim 8, wherein the pneumatic connector is a quick connector.

10. The reusable composite curing system of claim 8, wherein the end fitting is formed of a metallic material.

11. The reusable composite curing system of claim 8, wherein the passage runs perpendicular to the first half of the pneumatic connector.

12. A method of curing a composite material, the method comprising: using a reusable composite curing system for completing the following steps: lowering a reusable bladder towards a cure tool such that a first half of a pneumatic connector secured in an end fitting sealed to the reusable bladder connects to a second half of the pneumatic connector secured in the cure tool, the pneumatic connector forming a pressure passageway to provide an external pressure to an interior of the reusable bladder; and curing the composite material to form a composite part having an internal cavity by applying heat and pressure to the composite material supported on the cure tool and the reusable bladder, wherein the reusable composite curing system comprises: the reusable bladder; the end fitting sealed to the reusable bladder, wherein the end fitting comprises: a bore that is threaded, penetrates partially into the end fitting, and connects, at an interior most point of the bore, to a passage that exits the bore perpendicular to an axis of the bore and into communication with an interior of the reusable bladder; and the first half of the pneumatic connector threaded into the bore and in communication with the passage; and the cure tool that comprises a venting channel that extends from a supporting surface of the cure tool and exits through one of an underside or a side of the cure tool; and the second half of the pneumatic connector secured within and extending out from the venting channel, wherein the first half of the pneumatic connector is configured to release from the second half of the pneumatic connector while the second half of the pneumatic connector remains in the cure tool and while the first half of the pneumatic connector remains in the end fitting.

13. The method of claim 12 further comprising: sealing a reusable vacuum bag to the cure tool and over the end fitting and the reusable bladder.

14. The method of claim 12 further comprising: pressurizing the reusable bladder by subjecting the interior of the reusable bladder to the external pressure by way of the pressure passageway.

15. The method of claim 12 further comprising: wrapping the composite material around the reusable bladder; and wherein lowering the reusable bladder comprises placing the composite material and reusable bladder onto the cure tool.

16. The method of claim 12 further comprising: placing the composite material onto the cure tool;

and wherein lowering the reusable bladder towards the cure tool comprises positioning the reusable bladder onto the composite material on the cure tool.

17. The method of claim 12 further comprising: applying the composite material over the reusable bladder on the cure tool after lowering the reusable bladder towards the cure tool.

18. The method of claim 12 further comprising: removing the reusable bladder from the internal cavity of the composite part; and curing a second composite material to form a second composite part using the reusable bladder and the cure tool.
